



REPORT No.: SZ23020267E01

TEST REPORT

APPLICANT : Seeed Technology Co., Ltd.

PRODUCT NAME : reComputer J401 Carrier Board

MODEL NAME : J401

BRAND NAME : Seeed Studio

STANDARD(S) : 47 CFR Part 15 Subpart B

RECEIPT DATE : 2023-02-22

TEST DATE : 2023-03-03 to 2023-03-10

ISSUE DATE : 2023-03-16



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MORLAB

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DIRECTORY

1. Technical Information	3
1.1. Applicant and Manufacturer Information	3
1.2. Equipment Under Test (EUT) Description	3
2. Test Results	4
2.1. Applied Reference Documents	4
2.2. EUT Setup and Operating Conditions	5
3. 47 CFR Part 15B Requirements	6
3.1. Conducted Emission	6
3.2. Radiated Emission	10
Annex A Photographs of Test Setup	17
Annex B Photos of the EUT	19
Annex C Test Uncertainty	22
Annex D Testing Laboratory Information	23

Change History		
Version	Date	Reason for change
1.0	2023-03-16	First edition



1. Technical Information

Note: Provide by applicant.

1.1. Applicant and Manufacturer Information

Applicant:	Seed Technology Co., Ltd.
Applicant Address:	9F, G3 Building, TCL International E City, Zhongshanyuan Road, Nanshan District Shenzhen N/A China
Manufacturer:	Shenzhen Xinxian Technology Co., Limited
Manufacturer Address:	F5, Building B17, Hengfeng Industrial City, No. 739 Zhoushi Rd, Baoan District, Shenzhen, Guangdong, P.R.C. F5, Building B17, Hengfeng Industrial City, No. 739 Zhoushi Rd, Baoan District, Shenzhen, Guangdong, P.R.C.

1.2. Equipment Under Test (EUT) Description

Product Name:	reComputer J401 Carrier Board
EUT No.:	1#
Hardware Version:	V1.0
Software Version:	V35.2.1+Jetpack5.1

Note:

1. This test report is variant from the original report (Report No. : SZ23020264E01, model name: J4012) based on the similarity between before, J401 is the PCBA of J4012 which take away the enclosure, fan, Nvidia Orin NX. This test has been tested RE and CE. For RE and CE, the test results in this report still refer to the test results of the original test report. The others are the same as before.
2. For a more detailed description, please refer to specification or user's manual supplied by the applicant and/or manufacturer.



2. Test Results

2.1. Applied Reference Documents

The objective of the report is to perform testing according to 47 CFR Part 15 Subpart B:

No.	Identity	Document Title
1	47 CFR Part 15	Radio Frequency Devices

Test detailed items/section required by FCC rules and results are as below:

No.	Section	Description	Test Date	Test Engineer	Result	Method Determination Remark
1	15.107	Conducted Emission	2023.03.03	Wu Zhaoling	PASS ^{Note 4}	No deviation
2	15.109	Radiated Emission	2023.03.10	Yang Lian	PASS ^{Note 4}	No deviation

Note 1: The tests were performed according to the method of measurements prescribed in ANSI C63.4-2014.

Note 2: Additions to, deviation, or exclusions from the method shall be judged in the "method determination" column of add, deviate or exclude from the specific method shall be explained in the "Remark" of the above table.

Note 3: When the test result is a critical value, we will use the measurement uncertainty give the judgment result based on the 95% confidence intervals.

Note 4: The test results of these test items in this report refer to the test report (Report No.: SZ23020264E01).



2.2. EUT Setup and Operating Conditions

Test Item	
Radiated Emission	
Mode 1	: EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working
Conducted Emission	
Mode 1	: EUT + Adapter + Display Screen + Display Adapter + Mobile Phone + USB Flash Disk + RJ45 Link + Working

During the measurement, the environmental conditions were within the listed ranges:

Temperature (°C):	15 - 35
Relative Humidity (%):	30 - 60
Atmospheric Pressure (kPa):	86 - 106

3. 47 CFR Part 15B Requirements

3.1. Conducted Emission

3.1.1. Requirement

According to FCC section 15.107, the radio frequency voltage that is conducted back onto the AC power line on any frequency within the band 150kHz to 30MHz shall not exceed the limits in the following table, as measured using a 50 μ H/50 Ω line impedance stabilization network (LISN).

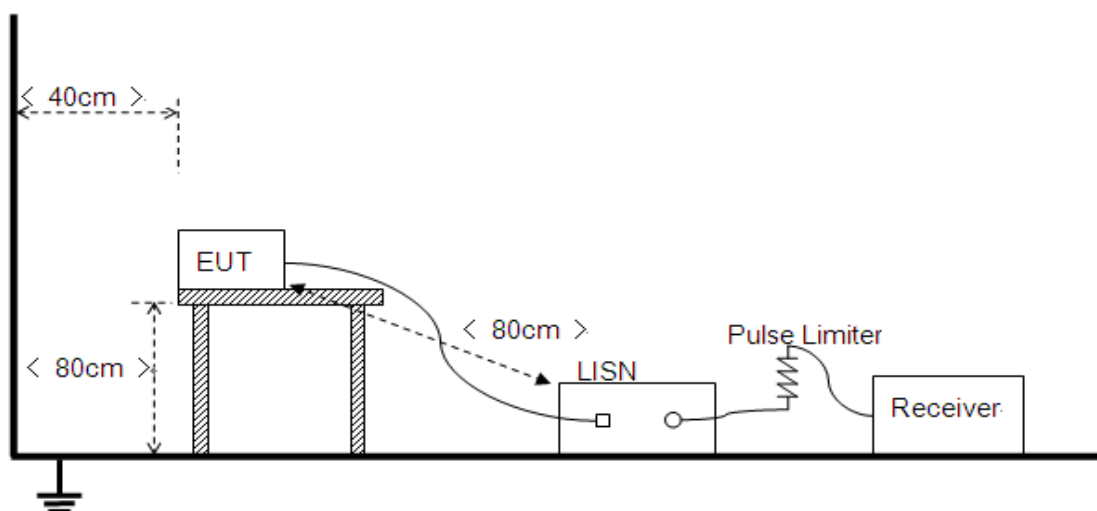
Frequency Range (MHz)	Conducted Limit (dB μ V)	
	Quasi-peak	Average
0.15 - 0.50	66 to 56	56 to 46
0.50 - 5	56	46
5 - 30	60	50

Note:

- The limit subjects to the Class B digital device.
- The lower limit shall apply at the band edges.
- The limit decreases linearly with the logarithm of the frequency in the range 0.15 - 0.50MHz.

3.1.2. Test Setup

Please refer to Annex A for the photographs of the Test Configuration.





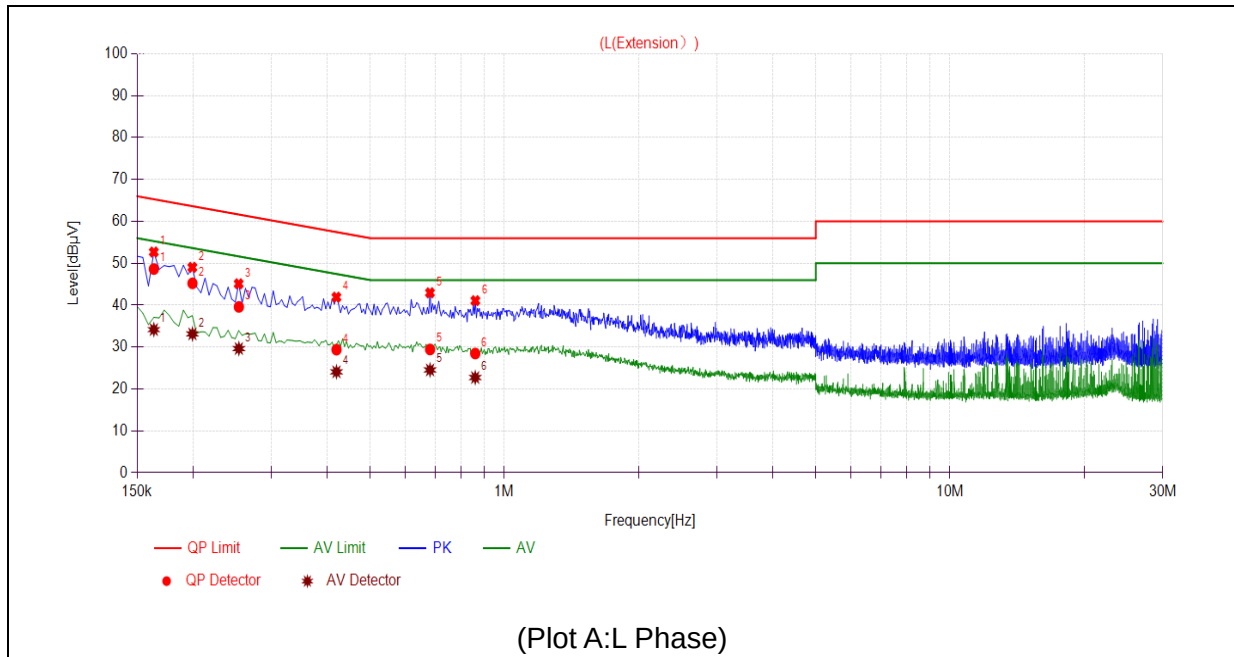
The EUT is placed on a 0.8m high insulating table, which stands on the grounded conducting floor, and keeps 0.4m away from the grounded conducting wall. The EUT is connected to the power mains through a LISN which provides $50\Omega/50\mu\text{H}$ of coupling impedance for the measuring instrument. A Pulse Limiter is used to protect the measuring instrument. The factors of the whole test system are calibrated to correct the reading.

The power strip or extension cord has been investigated to make sure that the LISN integrity is maintained with respect to the impedance characteristics as prescribed in ANSI C63.4-2014 at Clause 4.3.

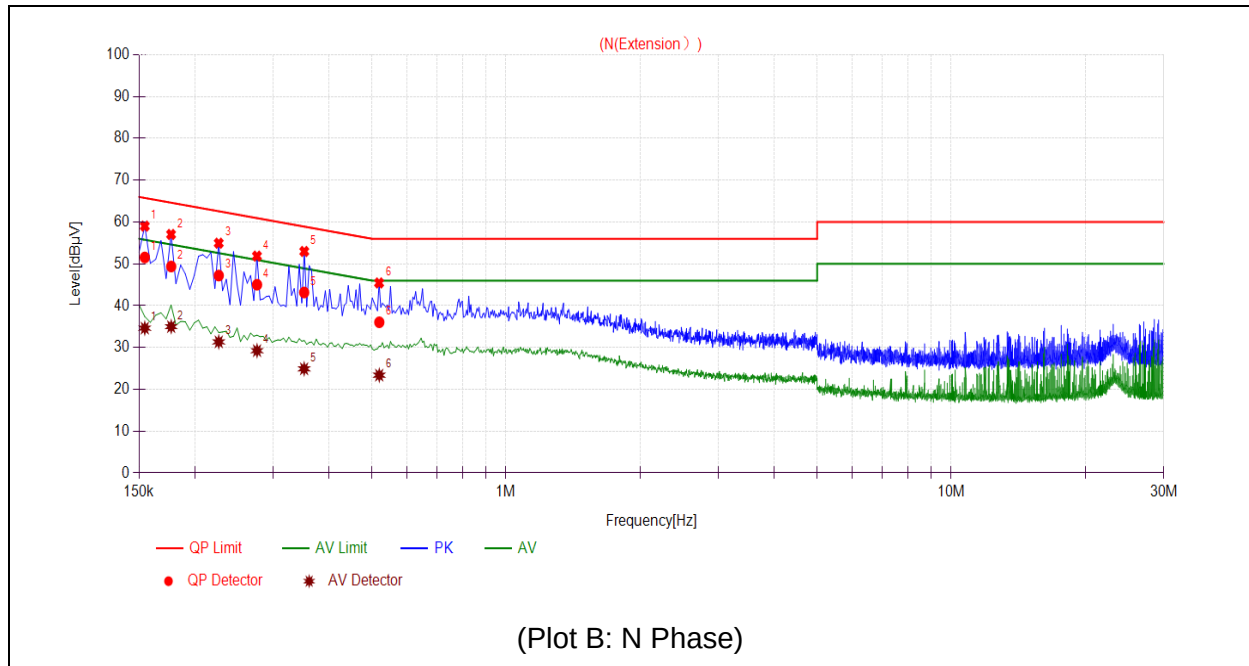
3.1.3. Test Result

Set RBW=9 kHz, VBW=30 kHz. The maximum conducted interference is searched using Peak (PK), Quasi-peak (QP) and Average (AV) detectors; the emission levels more than the AV and QP limits, and that have narrow margins from the AV and QP limits will be re-measured with AV and QP detectors. Tests for both L phase and N phase lines of the power mains connected to the EUT are performed. All test modes are considered, refer to recorded points and plots below.

A. Test Plot and Suspicious Points:



No.	Fre. (MHz)	Emission Level (dBμV)		Limit (dBμV)		Power-line	Verdict
		Quasi-peak	Average	Quasi-peak	Average		
1	0.1635	48.62	34.17	65.28	55.28	Line	PASS
2	0.1996	45.22	33.14	63.63	53.63		PASS
3	0.2537	39.59	29.64	61.63	51.63		PASS
4	0.4201	29.39	24.09	57.45	47.45		PASS
5	0.6813	29.43	24.54	56.00	46.00		PASS
6	0.8603	28.47	22.73	56.00	46.00		PASS



No.	Fre. (MHz)	Emission Level (dBμV)		Limit (dBμV)		Power-line	Verdict
		Quasi-peak	Average	Quasi-peak	Average		
1	0.1545	51.53	34.63	65.75	55.75	Neutral	PASS
2	0.1771	49.37	34.99	64.62	54.62		PASS
3	0.2265	47.21	31.35	62.58	52.58		PASS
4	0.2761	45.02	29.25	60.93	50.93		PASS
5	0.3522	43.16	24.91	58.91	48.91		PASS
6	0.5194	36.04	23.38	56.00	46.00		PASS



3.2. Radiated Emission

3.2.1. Requirement

According to FCC section 15.109 (a), the field strength of radiated emissions from unintentional radiators at a distance of 3 meters shall not exceed the following values:

Frequency Range (MHz)	Field Strength Limitation at 3m Measurement Dist	
	($\mu\text{V/m}$)	(dB $\mu\text{V/m}$)
30.0 - 88.0	100	20log 100
88.0 - 216.0	150	20log 150
216.0 - 960.0	200	20log 200
Above 960.0	500	20log 500

As shown in FCC section 15.35(b), for frequencies above 1000MHz, the field strength limits are based on average detector. When average radiated emission measurements are specified in this part, including emission measurements below 1000MHz, there also is a limit on the radio frequency emissions, as measured using instrumentation with a peak detector function, corresponding to 20dB above the maximum permitted average limit for the frequency being investigated unless a different peak emission limit is otherwise specified in the rules.

Note:

- 1) The tighter limit shall apply at the boundary between two frequency range.
- 2) Limitation expressed in dB $\mu\text{V/m}$ is calculated by $20\log$ Emission Level($\mu\text{V/m}$).

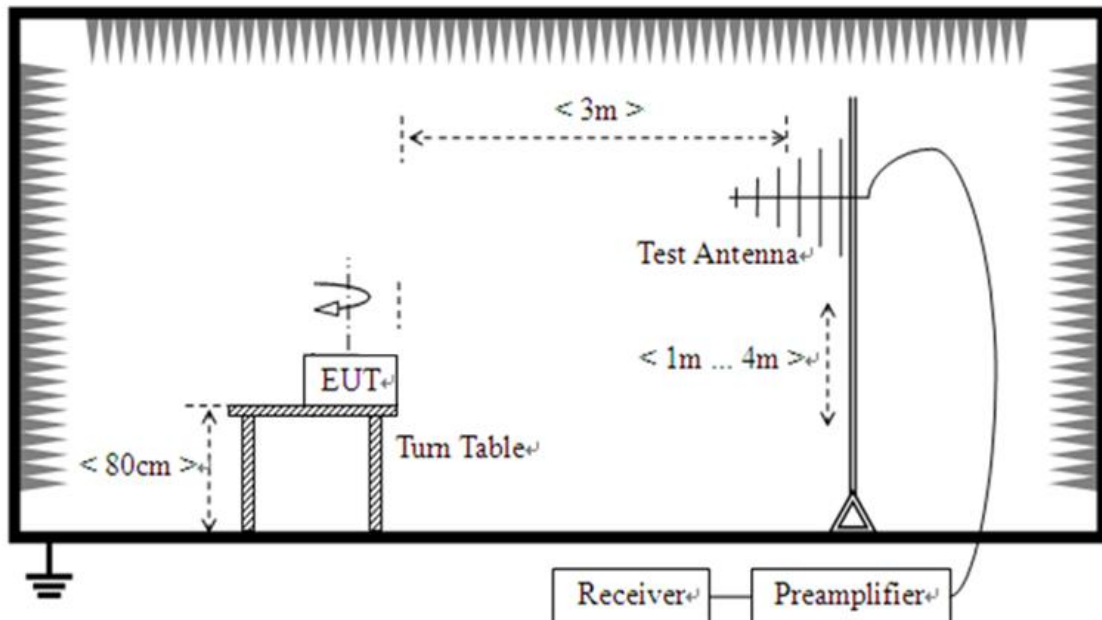
3.2.2. Frequency Range of Measurement

According to 15.33(b)(1), the frequency range of radiated measurement for the EUT is listed in the following table:

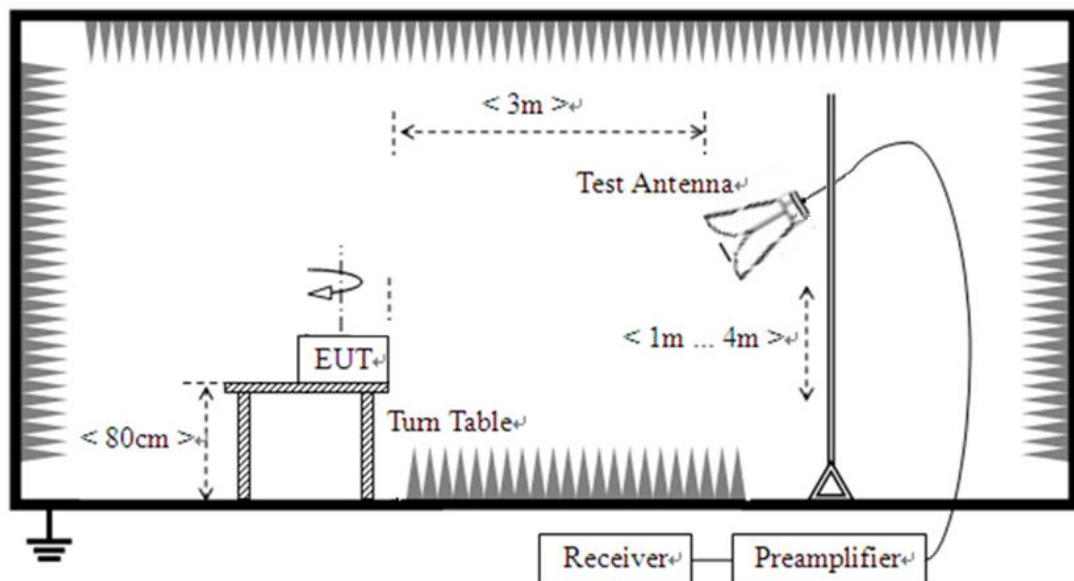
Highest frequency generated or used in the device or on which the device operates or tunes (MHz)	Upper frequency of measurement range (MHz)
Below 1.705	30.
1.705–108	1000.
108–500	2000.
500–1000	5000.
Above 1000	5th harmonic of the highest frequency or 40 GHz, whichever is lower.

3.2.3. Test Setup

- 1) For radiated emissions from 30MHz to 1GHz



- 2) For radiated emissions above 1GHz





The test is performed in a 3m Semi-Anechoic Chamber; the antenna factor, cable loss and so on of the site (factors) is calculated to correct the reading. The EUT is placed on a 0.8m high insulating Turn Table, and keeps 3m away from the Test Antenna, which is mounted on a variable-height antenna master tower.

For the test Antenna:

In the frequency range above 30MHz, Bi-Log Test Antenna (30MHz to 1GHz) and Horn Test Antenna (above 1GHz) are used. Test Antenna is 3m away from the EUT. Test Antenna height is varied from 1m to 4m above the ground to determine the maximum value of the field strength. The emission levels at both horizontal and vertical polarizations should be tested. For measurements above 1 GHz, keeping the measurement antenna aimed at the source of emissions at each frequency of significant emissions, with polarization oriented for maximum response.

For measurements below 1GHz the resolution bandwidth is set to 120 kHz for peak detection measurements or 120kHz for quasi-peak detection measurements. Peak detection is used unless otherwise noted as quasi-peak.

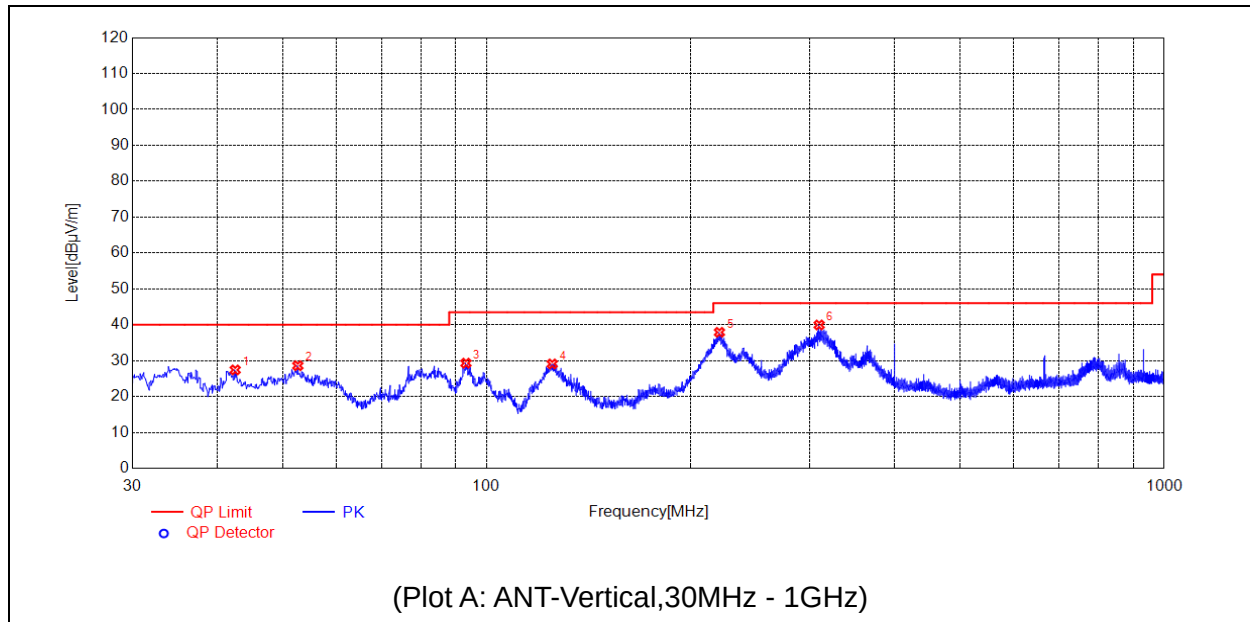
For measurements above 1GHz the resolution bandwidth is set to 1MHz, the video bandwidth is set to 3MHz for peak measurements and as applicable for average measurements.

3.2.4. Test Result

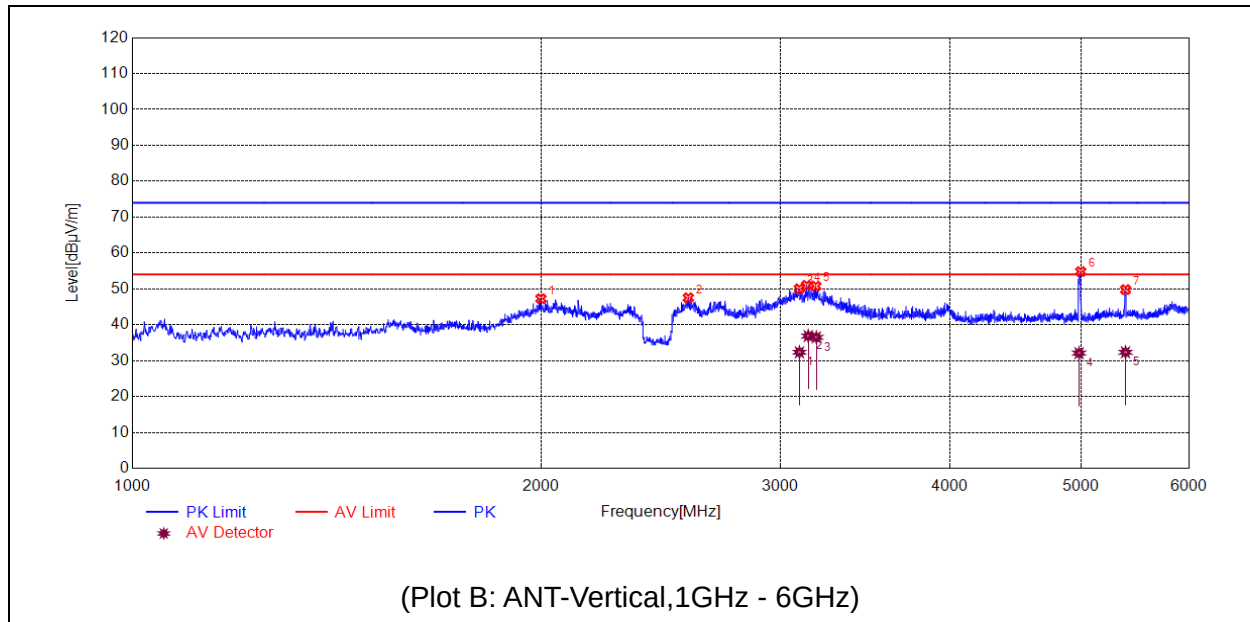
The maximum radiated emission is searched using PK, QP and AV detectors; the emission levels more than the limits, and that have narrow margins from the limits will be re-measured with AV and QP detectors. Both the vertical and the horizontal polarizations of the Test Antenna are considered to perform the tests. All test modes are considered, refer to recorded points and plots below.

The amplitude of emissions which are attenuated more than 20 dB below the permissible value need not be reported.

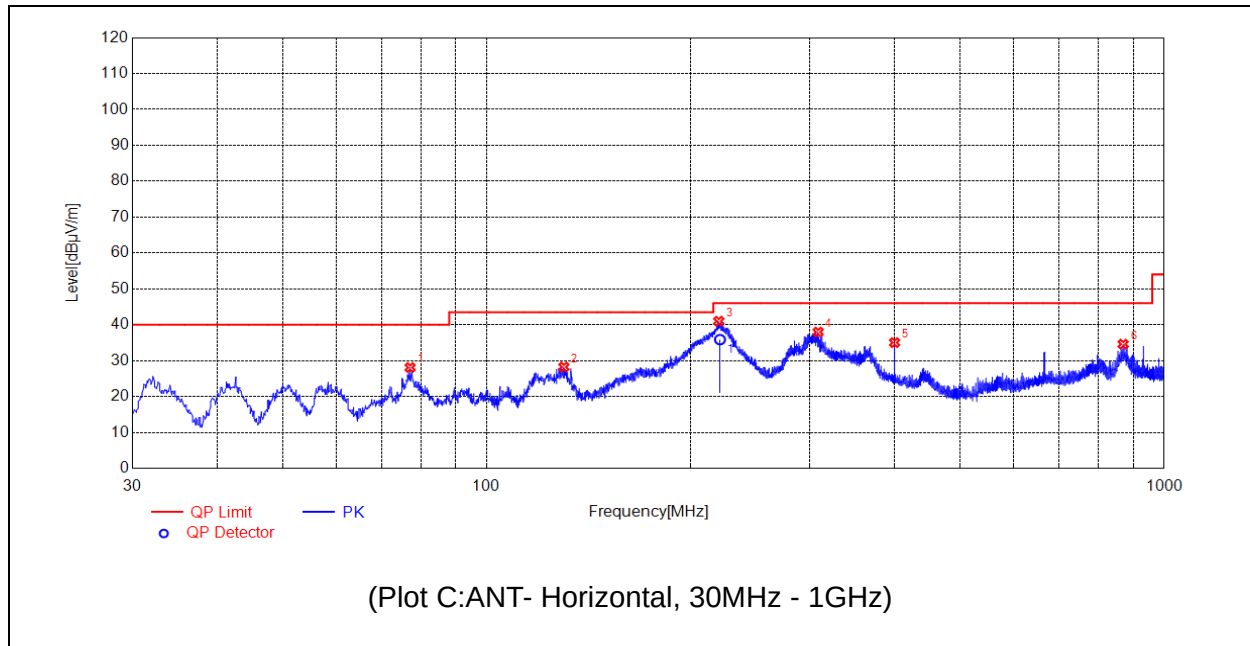
Note: All radiated emission tests were performed in X, Y, Z axis direction, and only the worst axis test condition was recorded in this test report.



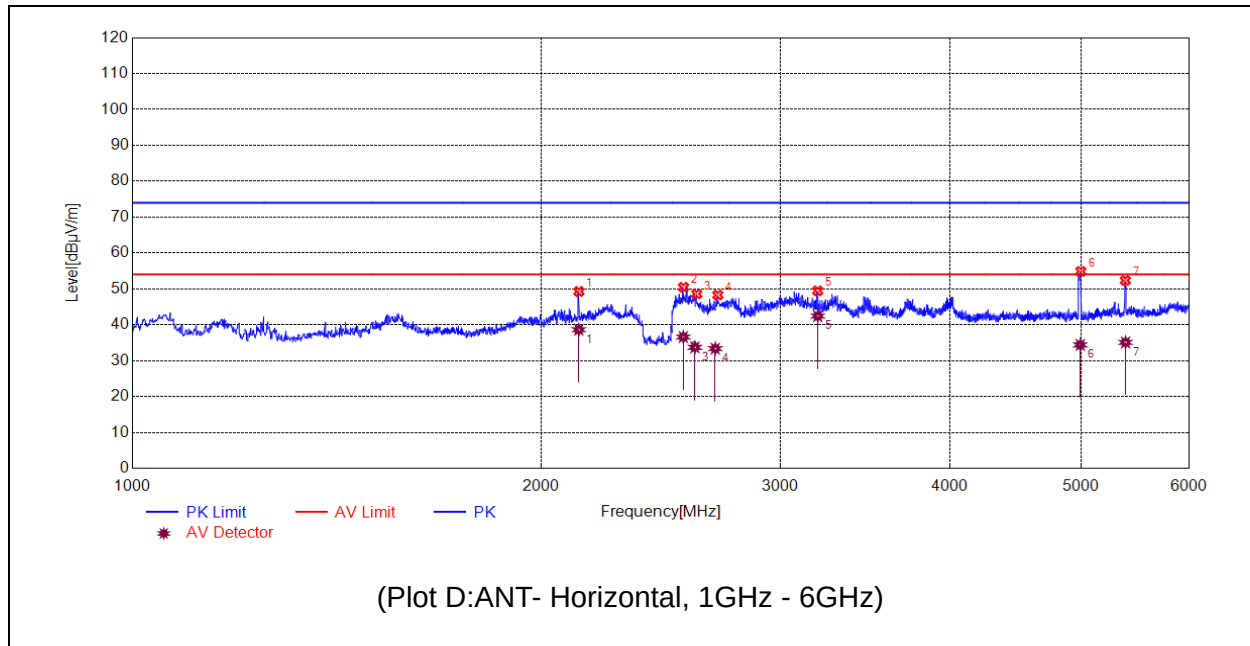
No.	Fre. MHz	PK dBμV/m	QP dBμV/m	AV dBμV/m	Limit-PK dBμV/m	Limit-QP dBμV/m	Limit-AV dBμV/m	ANT	Verdict
1	42.5143	27.42	N.A.	N.A.	N.A.	40.00	N.A.	V	PASS
2	52.6033	28.59	N.A.	N.A.	N.A.	40.00	N.A.	V	PASS
3	93.1533	29.30	N.A.	N.A.	N.A.	43.50	N.A.	V	PASS
4	124.8755	29.08	N.A.	N.A.	N.A.	43.50	N.A.	V	PASS
5	220.5271	37.90	N.A.	N.A.	N.A.	46.00	N.A.	V	PASS
6	309.4849	39.94	N.A.	N.A.	N.A.	46.00	N.A.	V	PASS



No.	Fre. MHz	PK dBμV/m	QP dBμV/m	AV dBμV/m	Limit-PK dBμV/m	Limit-QP dBμV/m	Limit-AV dBμV/m	ANT	Verdict
1	1999.1998	47.26	N.A.	N.A.	74.00	N.A.	54.00	V	PASS
2	2567.3135	47.52	N.A.	N.A.	74.00	N.A.	54.00	V	PASS
3	3099.4199	50.02	N.A.	32.34	74.00	N.A.	54.00	V	PASS
4	3138.4277	51.00	N.A.	36.85	74.00	N.A.	54.00	V	PASS
5	3187.4375	50.74	N.A.	36.49	74.00	N.A.	54.00	V	PASS
6	4995.7992	54.82	N.A.	32.10	74.00	N.A.	54.00	V	PASS
7	5390.8782	49.80	N.A.	32.33	74.00	N.A.	54.00	V	PASS



No.	Fre. MHz	PK dBμV/m	QP dBμV/m	AV dBμV/m	Limit-PK dBμV/m	Limit-QP dBμV/m	Limit-AV dBμV/m	ANT	Verdict
1	77.0497	28.09	N.A.	N.A.	N.A.	40.00	N.A.	H	PASS
2	129.9200	28.24	N.A.	N.A.	N.A.	43.50	N.A.	H	PASS
3	220.0420	41.00	35.87	N.A.	N.A.	46.00	N.A.	H	PASS
4	308.4178	37.97	N.A.	N.A.	N.A.	46.00	N.A.	H	PASS
5	399.8980	35.03	N.A.	N.A.	N.A.	46.00	N.A.	H	PASS
6	869.6190	34.60	N.A.	N.A.	N.A.	46.00	N.A.	H	PASS



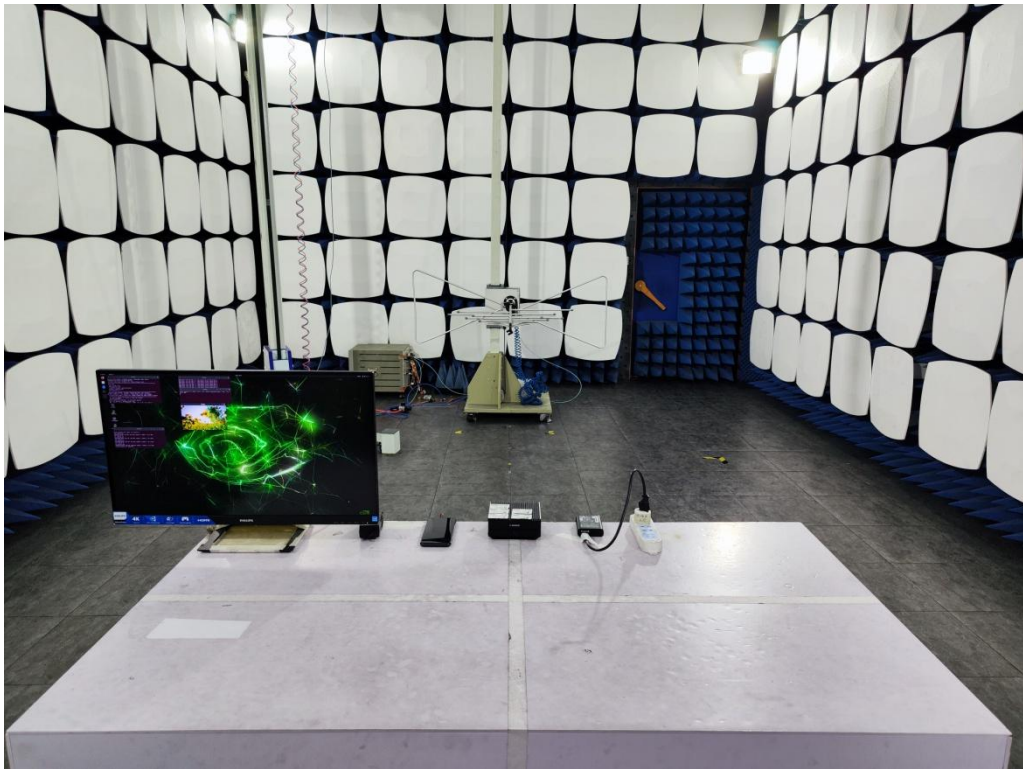
No.	Fre. MHz	PK dBμV/m	QP dBμV/m	AV dBμV/m	Limit-PK dBμV/m	Limit-QP dBμV/m	Limit-AV dBμV/m	ANT	Verdict
1	2132.2264	49.28	N.A.	38.63	74.00	N.A.	54.00	H	PASS
2	2546.3093	50.46	N.A.	36.58	74.00	N.A.	54.00	H	PASS
3	2604.3209	48.58	N.A.	33.71	74.00	N.A.	54.00	H	PASS
4	2699.3399	48.29	N.A.	33.35	74.00	N.A.	54.00	H	PASS
5	3197.4395	49.46	N.A.	42.41	74.00	N.A.	54.00	H	PASS
6	4996.7994	54.87	N.A.	34.39	74.00	N.A.	54.00	H	PASS
7	5388.8778	52.35	N.A.	35.09	74.00	N.A.	54.00	H	PASS

Annex A Photographs of Test Setup

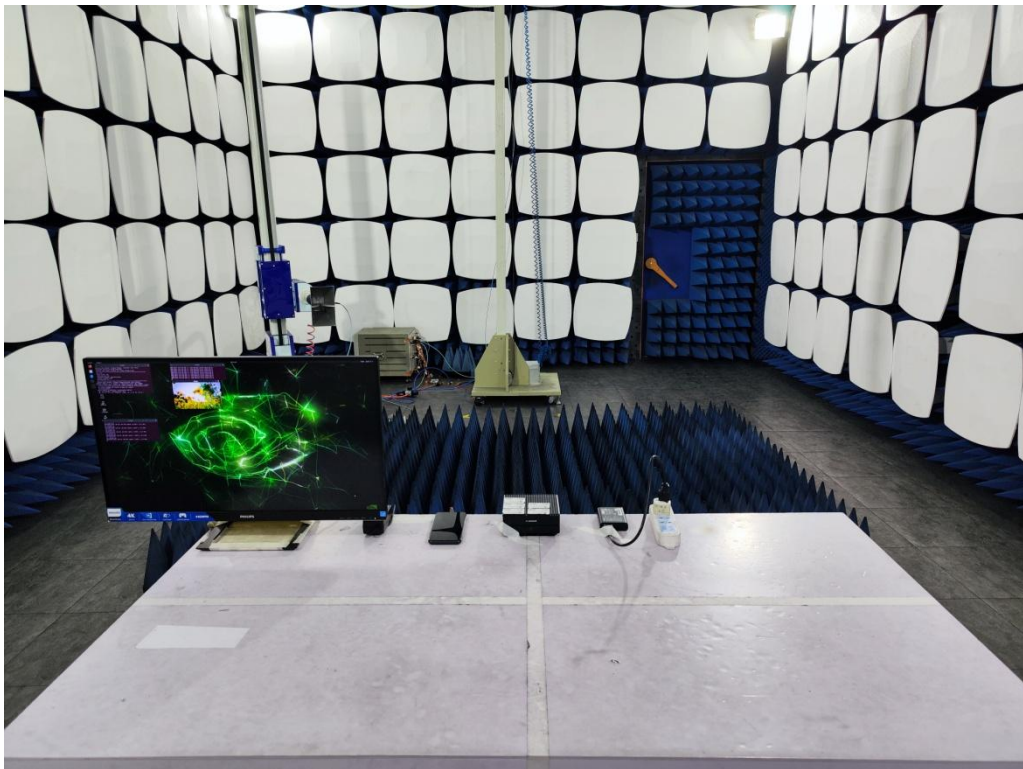
1. Conducted Emission



2. Radiated Emission(30MHz-1GHz)

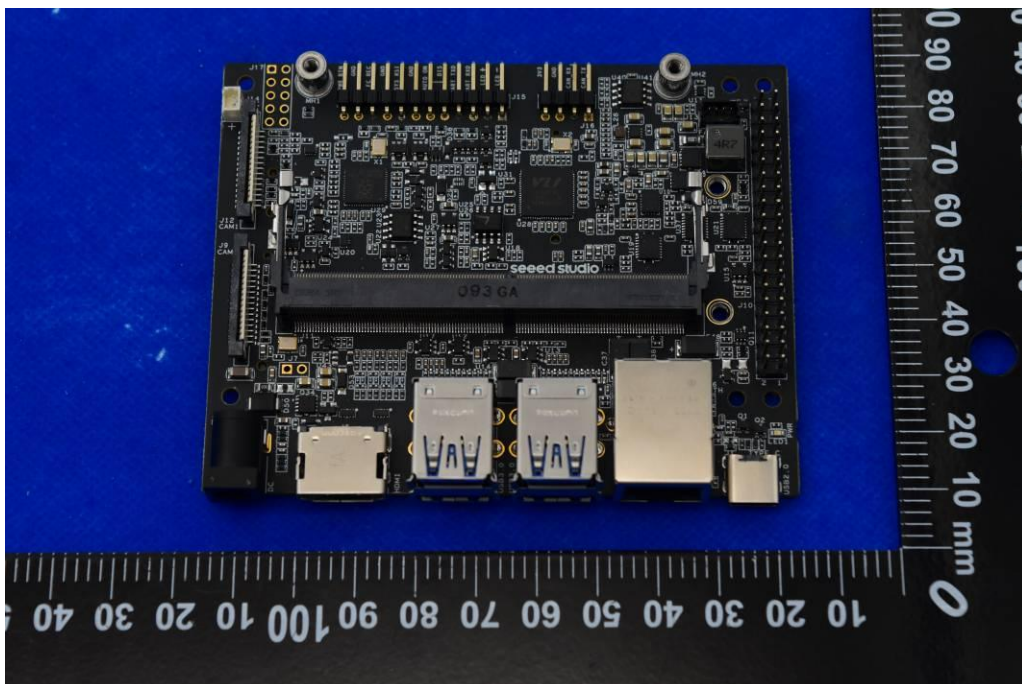
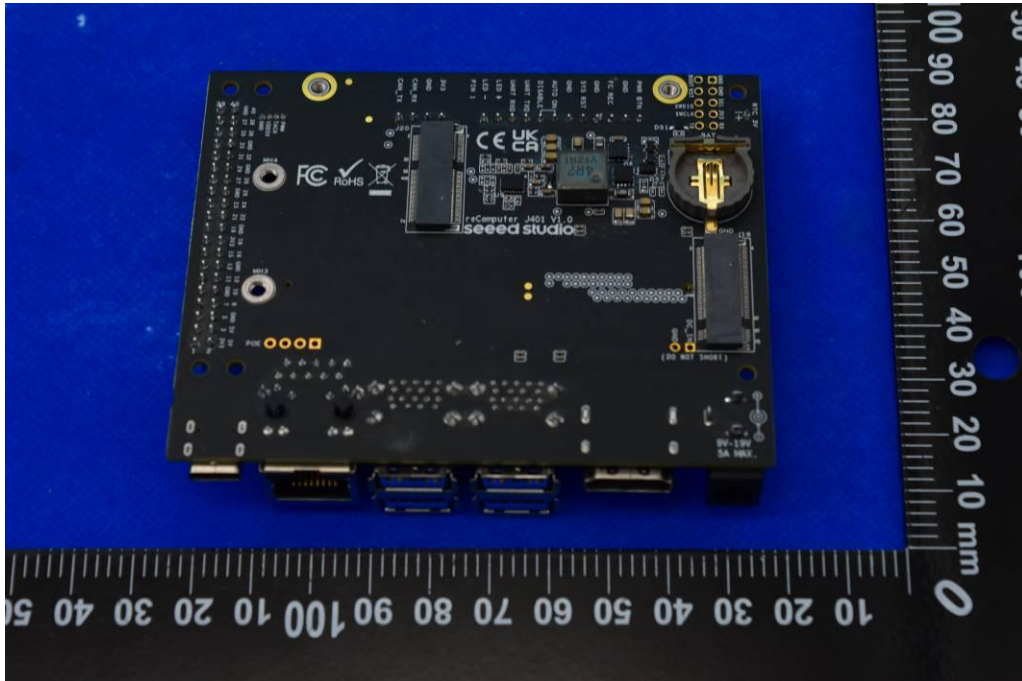


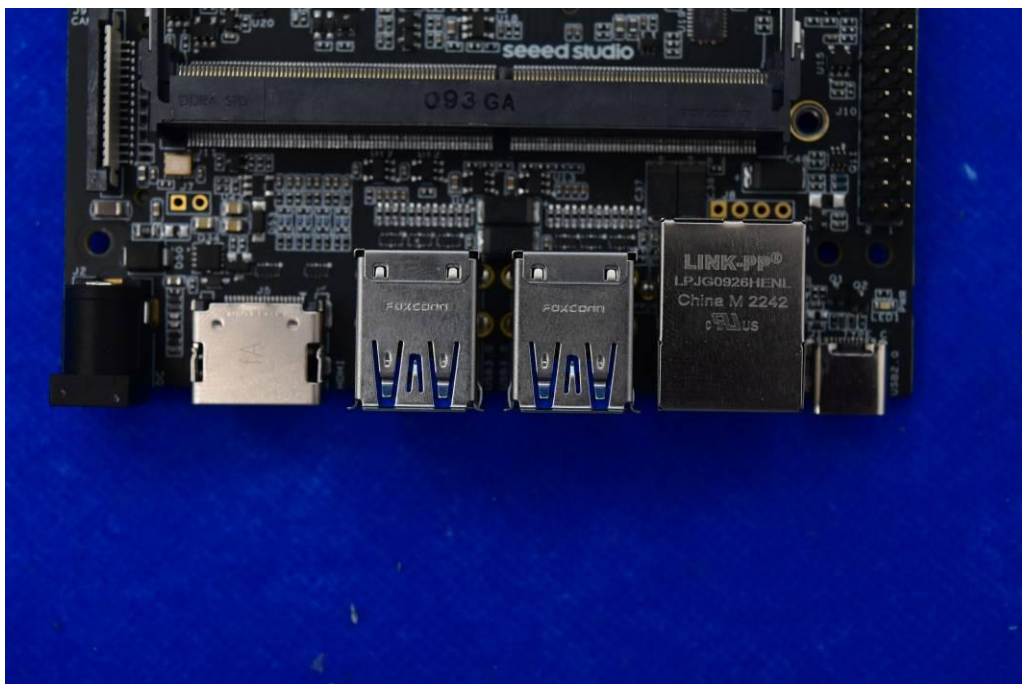
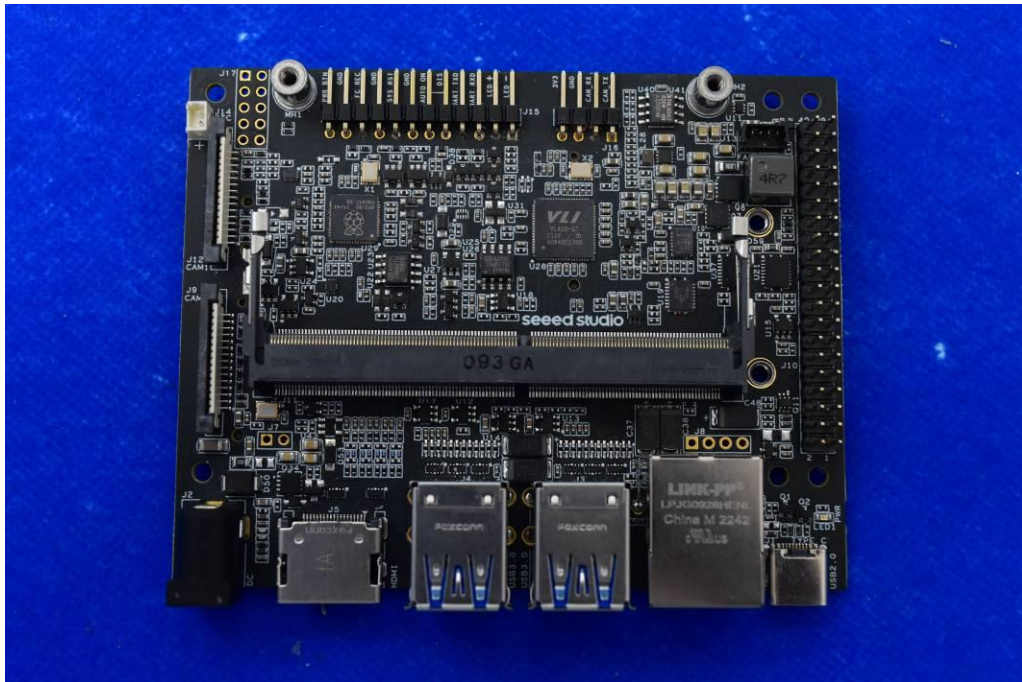
3. Radiated Emission(above 1GHz)

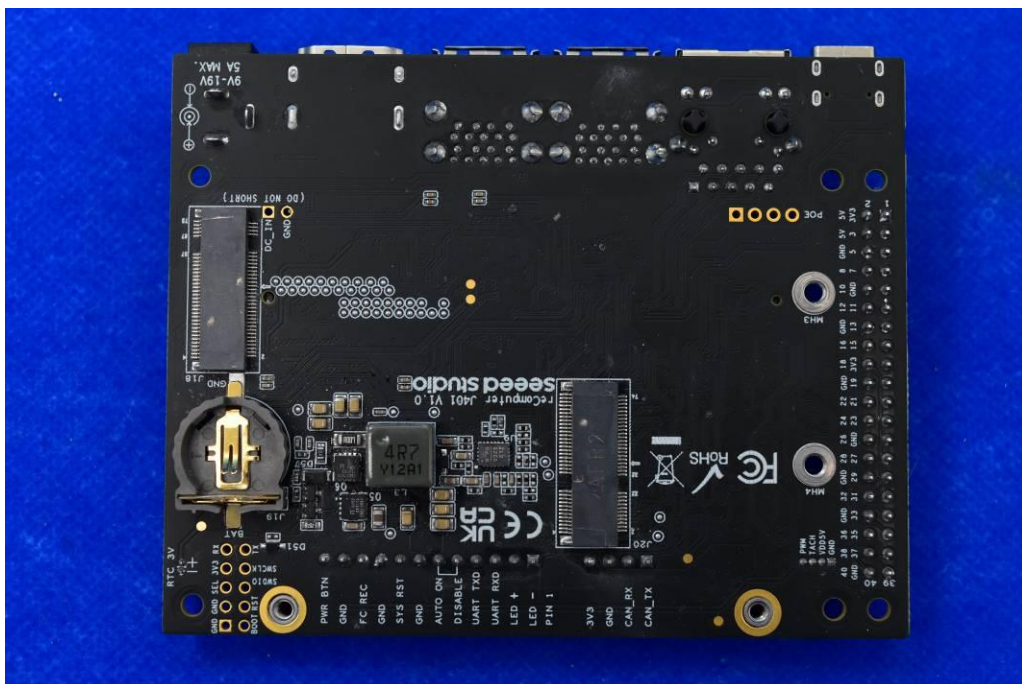
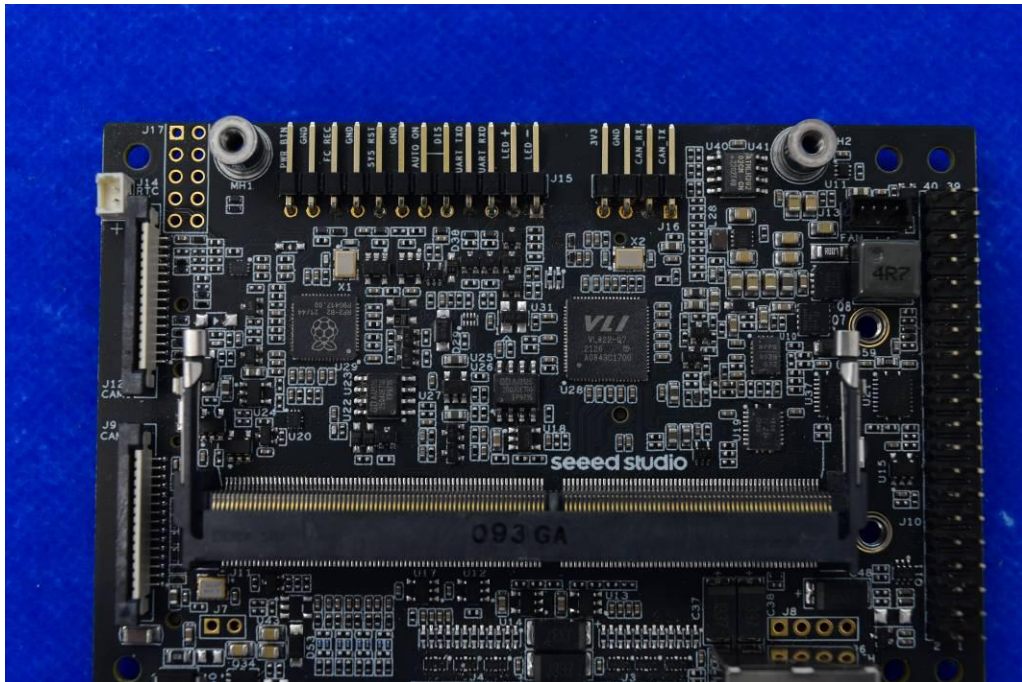


Annex B Photos of the EUT

1. Appearance of the EUT







Annex C Test Uncertainty

The uncertainty is calculated using the methods suggested in the "Guide to the Expression of Uncertainty in Measurement" (GUM) published by ISO.

Uncertainty of Conducted Emission Measurement

Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))	9kHz-150kHz	±3.3dB
	150kHz-30MHz	±2.8dB

Uncertainty of Radiated Emission Measurement

Measuring Uncertainty for a Level of Confidence of 95%(U=2Uc(y))	30MHz-200MHz	±5.06dB
	200MHz-1000MHz	±5.04dB
	1GHz-6GHz	±5.18dB
	6GHz-18GHz	±5.48dB



Annex D Testing Laboratory Information

1. Identification of the Responsible Testing Laboratory

Laboratory Name:	Shenzhen Morlab Communications Technology Co., Ltd.
Laboratory Address:	FL.3, Building A, FeiYang Science Park, No.8 LongChang Road, Block 67, BaoAn District, ShenZhen, GuangDong Province, P. R. China
Telephone:	+86 755 36698555
Facsimile:	+86 755 36698525

2. Identification of the Responsible Testing Location

Name:	Shenzhen Morlab Communications Technology Co., Ltd.
Address:	FL.3, Building A, FeiYang Science Park, No.8 LongChang Road, Block 67, BaoAn District, ShenZhen, GuangDong Province, P. R. China

3. Accreditation Certificate

Accredited Testing Laboratory:	The FCC designation number is CN1192. Test firm registration number is 226174. (Shenzhen Morlab Communications Technology Co., Ltd.)
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4. Test Software Utilized

Model	Version Number	Producer
TS+ -[JS32-RE]	Version 2.5.0.6	Tonscend
TS+ -[JS32-CE]	Version 2.5.0.0	Tonscend

**5. Test Equipments Utilized**

Description	Model	Serial No.	Manufacturer	Cal. Date	Due. Date
Bi-Log Antenna	VULB 9163	9163-274	SCHWARZBECK	2022/11/7	2025/11/6
Bi-Log Antenna	VULB 9163	9163-519	SCHWARZBECK	2022/5/25	2025/5/24
Horn Antenna	BBHA 9120D	9120D-963	SCHWARZBECK	2022/5/25	2025/5/24
Horn Antenna	BBHA 9120D	01774	SCHWARZBECK	2022/7/13	2025/7/12
Horn Antenna	BBHA9170	BBHA9170#773	SCHWARZBECK	2022/7/14	2025/7/13
Receiver	N9038A	MY54130016	Agilent	2022/7/7	2023/7/6
Receiver	N9038A	MY56400093	KEYSIGHT	2023/2/9	2024/2/8
6db Attenuator	BW-N6W5+	E191001	Mini-circuits	2022/10/11	2023/10/10
Preamplifier	S020180L3 203	61171/61172	LUCIX CORP.	2022/7/8	2023/7/7
Preamplifier	S10M100L3 802	46732	LUCIX CORP.	2022/7/8	2023/7/7
Preamplifier	DCLNA011 8-40C-S	DS77209	Decentest	2022/7/23	2023/7/22
RF Coaxial Cable	PE330	MRE001	Pasternack	N/A	N/A
RF Coaxial Cable	CLU18	MRE002	Pasternack	N/A	N/A
RF Coaxial Cable	CLU18	MRE003	Pasternack	N/A	N/A
RF Coaxial Cable	QA360-40-KK-0.5	22290045	Qualwave	N/A	N/A
RF Coaxial Cable	QA360-40-KKF-2	22290046	Qualwave	N/A	N/A
RF Coaxial Cable	QA500-18-NN-5	22120181	Qualwave	N/A	N/A
RF Coaxial Cable	BNC	MRE04	Qualwave	N/A	N/A
Receiver	ESPI	101052	R&S	2022/7/7	2023/7/3
LISN	NSLK 8127	8127449	Schwarzbeck	2023/2/21	2024/2/20



REPORT No.: SZ23020267E01

10dB Pulse Limiter	VTSD 9561-F	VTSD 9561 F-B #206	SCHWARZBECK	2022/7/6	2023/7/5
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6. Ancillary Equipment Utilized

Description	Manufacturer	Model	Serial No.
Display Adapter	JIELIAN	ADPC2065	Q40G325B-615-01I ZA
Display Screen	Gaokeshuma	GN-2019001	N/A
Mobile Phone	REALME	RMX3193	VGSKUKT85TLFBMPJ
USB Flash Disk	SanDisk	MSIP-REM-TAD-SDCZ73	BL210726210z
Adapter	LITEON	HA-1600-12	N/A

_____ END OF REPORT _____